



Does control and target matter in the CZ (Controlled-Z) Gate? [duplicate]

Asked 3 years, 6 months ago Modified 3 years, 6 months ago Viewed 5k times

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[What is the control qubit in this 1D ladder of controlled Z gates?](#) (1 answer)

Closed 3 years ago.

IBM Quantum Experience and other Algorithm Creators generally draw the CZ gates like this:



Does it not matter which qubit is the control and which is the target? If so why?

quantum-gate quantum-control

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answered May 27, 2021 at 17:30
 Jedon Erwin
201 2 7

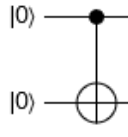
- 1 Have you reviewed the matrix for the CZ gate? – Mark Spradell May 27, 2021 at 17:32
- Yes but I'm trying to understand it intuitively. To me, I think of the CZ gate as "if (control qubit) flip phase of target qubit" similar to a CNOT, but obviously this isn't the case – Jedon Erwin May 27, 2021 at 17:45
- 4 Perhaps a better intuitive way to understand the CZ gate is: "flip the sign of the amplitude on the $|11\rangle$ state and do nothing to $|00\rangle$, $|01\rangle$ and $|10\rangle$ ". Mathematically, CZ maps $\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$ to $\alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle - \delta|11\rangle$. Clearly, the qubit roles are symmetric. – Adam Zakman May 27, 2021 at 18:24 ✓
- I think it is matter that the first qubit you put in $\text{czi}(\text{c})$ is the one that control the phase and the second qubit you put in is the target one. The picture which Qiskit illustrate is just a way to show the gate in circuit – BAQ BACH GUA May 27, 2021 at 18:26

1 Answer Sorted by: Highest score (default) 5

First, note that the Controlled-X gate can be written as:

$$CX = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$$

This tells us that the **first qubit is the controlled**, and the **second qubit is the target**. And when the controlled qubit is $|0\rangle$ we do nothing hence the Identity operator. When the controlled-qubit is a $|1\rangle$ we apply the X operator. So when you look at CX gate you see something like:



to indicate that the first qubit is the controlled and the second is the target since it matters. Also note that CX has the matrix representation in the computational basis as:

$$CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Now, if you look at the the controlled-Z (CZ) gate matrix representation in the computational basis, you will see:

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

But note that this can be written as:

$$CZ = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z = I \otimes |0\rangle\langle 0| + Z \otimes |1\rangle\langle 1|$$

what this tells us is that it's symmetric... it doesn't matter which qubit is the controlled or target. So why not just create a controlled gate with both being the controlled qubits.. that is why you see CZ usually shown as:



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answered May 27, 2021 at 18:20
 KAJ226
14.1k 2 11 33

Note that for this to work you need (the conventional) $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, contrary to what you'd usually have when you write a qubit Hamiltonian with σ_z . – Daniel Mar 7, 2022 at 21:16 ✓